MNYMO Girls

August 15, 2026

Individual

The answer key is below.

p	Answer	p	Answer
1	1337	6	682
2	950	7	0
3	$5 + \pi$	8	110
4	$\frac{1}{3}$	9	$\frac{107}{288}$
5	$\sqrt{3}$	10	$\frac{8400}{583}$

1. What is $\frac{7776}{6} + 6 \cdot 7 + 6 - 7$?

   Proposed by: Angie Huang

   Answer: $\boxed{1337}$

   Solution: We have $\frac{7776}{6} + 6 \cdot 7 + 6 - 7 = 1296 + 42 + 6 - 7 = 1337$.

2. Daniel, Michael, and Xindi are playing a game. Initially, Daniel, Michael, and Xindi have 1000 dollars. Each day, Xindi gives Michael 5 dollars, Michael gives Daniel 10 dollars, and Daniel gives Xindi 20 dollars. After ten days, how much money does Michael have?

   Proposed by: Michael Luo

   Answer: $\boxed{950}$

   Solution: Observe that Michael gains 5 dollars from Xindi each day and loses 10 dollars to Daniel each day. Hence he loses 5 dollars each day, so after ten days, Michael is down 50 dollars. Hence he has $1000 - 50 = \boxed{950}$ dollars.

3. A square is drawn with side length 1. Compute the area of the region of points with distance at most 1 from some point in the square.

   Proposed by: Michael Luo

   Answer: $\boxed{5 + \pi}$

   Solution: The region is a rounded circle which may be decomposed into five squares and four quarter-circles, with side length 1 and radius 1, respectively. So the answer is $5 \cdot 1^2 + \pi \cdot 1^2 = \boxed{5 + \pi}$.

4. Sneha has a deck of 10 cards labeled with positive integers 1 to 10. She draws them randomly from the deck one at a time. What is the probability that she draws both the 6 and 7 before the 8?

   Proposed by: Angie Huang

   Answer: $\boxed{\frac{1}{3}}$

   Solution: Imagine randomly ordering all 10 cards in a line. Let this be Sneha's drawing order. The desired probability is the same as the probability that the 8 is to the right of both the 6 and 7. This happens with probability $\frac{2!}{3!} = \frac{1}{3}$.

5. Triangle ABC has $AB = 6$, $AC = 10$, and $\angle BAC = 120^\circ$. What is its inradius?

   Proposed by: Angie Huang

   Answer: $\boxed{\sqrt{3}}$

   Solution: By the Law of Cosines, $BC^2 = 6^2 + 10^2 - 2 \cdot 6 \cdot 10 \cdot \cos 120^\circ = 196$, so $BC = 14$. Let the incircle, with center O, touch AB,AC, and BC at the points D, E, and F respectively. Since the semiperimeter is 15, we have $AD = 1$. Since $\angle DAO = 60^\circ$, this means the inradius has length $\sqrt{3}$.

6. If $x + \frac{1}{x} = 7$, what is the remainder when $x^6 + \frac{1}{x^6}$ is divided by 1000?

   Proposed by: Addison Shi

Answer: $\boxed{682}$

Solution: $x^2 + \frac{1}{x^2} = 47$, $x^4 + \frac{1}{x^4} = 2207$, $x^6 + \frac{1}{x^6} = 47 \cdot 2207 - 47 = 103682$.

7. What is the last digit of $1^1 + 2^2 + 3^3 + 4^4 + \cdots + 100^{100}$?

   Proposed by: Sneha Kundu

   Answer: $\boxed{0}$

   Solution:

   Observe that the last digits of these powers cycle $\bmod 4$. Further, the base of each term in the expression can be simplified to simply the units digit raised to the same exponent. In other words, we wish to find the last digit of $1^1 + 2^2 + \cdots + 1^{11} + 2^{12} + \cdots 0^{100}$. Due to the cycling, we only need to calculate $5(1^1 + 2^2 + 3^3 + \cdots 1^{11} + 2^{12} + \cdots 0^{20}) \pmod 10$. This simplifies to $24 \cdot 5 \equiv 0 \pmod 10$.

8. What is the sum of the 5 smallest positive integers that each have exactly 6 positive factors?

   Proposed by: Elizabeth Zhu

   Answer: $\boxed{110}$

   Solution: If a positive integer has six positive factors, its prime factorization must be either in the form p^5 or $p^2 \cdot q$, where p and q are distinct primes. After testing the smallest prime factorizations, the smallest five are found to be $2^2 \cdot 3 = 12$, $3^2 \cdot 2 = 18$, $2^2 \cdot 5 = 20$, $2^2 \cdot 7 = 28$, and $2^5 = 32$. So, the desired sum is $12 + 18 + 20 + 28 + 32 = 110$.

9. Angie and Sneha are playing pickleball and will arrive at a uniformly random time between 5 and 6 PM. Angie will wait for an interval of 10 minutes whereas Sneha will wait for an interval of 15 minutes. What is the probability that they will meet and play a match?

   Proposed by: Sneha Kundu

   Answer: $\boxed{\frac{107}{288}}$

   Solution: We can compute the probability by imagining this scenario geometrically. Consider the outcome space a $60x60$ square with the lower left corner at the origin. Then, Angie's time (in minutes after 5PM) can be represented by the y-axis. Sneha's time, measured in the same way, can be represnted by the x-axis. Then, the solution space is the strip bounded by the edges of the square, and the lines $y = x - 15$ and $y = x + 10$. The area of this strip is the area of the square minus the two triangles surrounding the strip. This is computed as

$$\frac{60^2 - (\frac{50^2}{2} + \frac{45^2}{2})}{60^2} = \frac{12^2 - (\frac{10^2}{2} + \frac{9^2}{2})}{12^2} = \frac{144 - (50 + 40.5)}{144} = \frac{107}{288}.$$

10. Let $\triangle ABC$ have sidelengths $AB = 13$, $BC = 14$, and $AC = 15$. If there is a point D on side BC such that $BD = 6$ and a point E on side AC such that $\frac{AE}{EC} = \frac{5}{6}$, let the intersection point of lines AD and BE be X. Find the area of $\triangle AXE$.

    Proposed by: Addison Shi

    Answer: $\boxed{\frac{8400}{583}}$

    Solution: We have $[ABC] = 84$. Then $[ABD] = 84 \cdot \frac{4}{7} = 48$. From mass points, $\frac{AX}{XD} = \frac{35}{18}$. $[AXC] = 48 \cdot \frac{35}{53}$. We compute $AXE = [AXC] \cdot \frac{5}{11} = \frac{8400}{583}$.